

CSE-421

Computer Graphics:

2D Transformations

Lecture-01

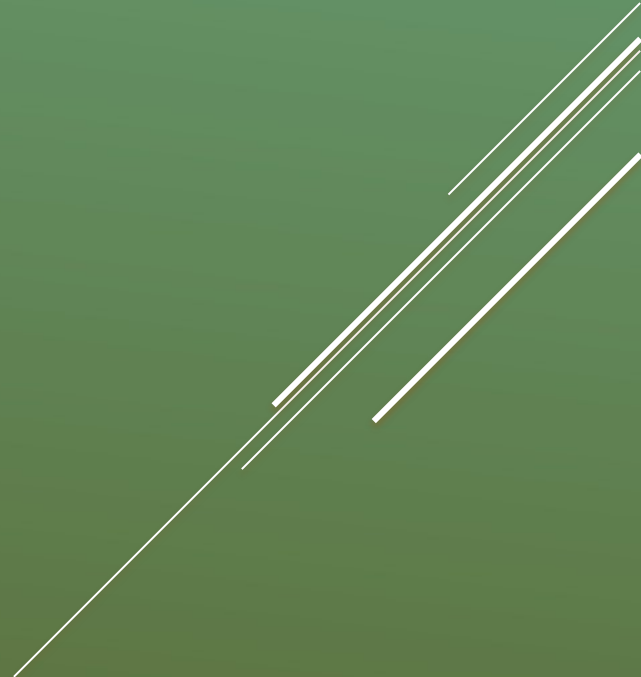
Lecturer, Varendra University

2D Transformations

“Transformations are the operations applied to geometrical description of an object to change its position, orientation, or size are called geometric transformations”.

Given a 2D object, transformation is to change the object's

- *Position (translation)*
- *Size (scaling)*
- *Orientation (rotation)*
- *Shapes (shear)*



Point Representation

- We can use a column vector (a 2x1 matrix) to represent a 2D point.



- Apply a sequence of matrix multiplications to the object vertices.

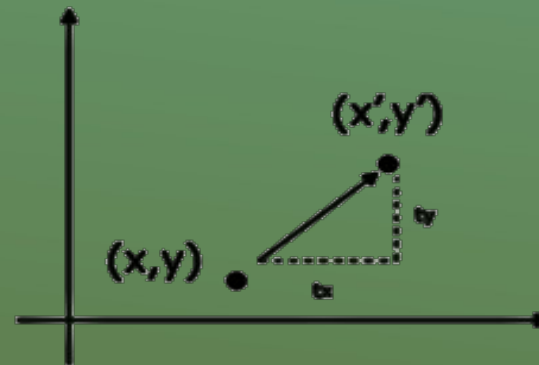
Translation

- Re-position a point along a straight line
- Given a point (x,y) , and the translation distance (t_x, t_y)

The new point, $P' : (x', y')$

$$x' = x + t_x$$

$$y' = y + t_y$$



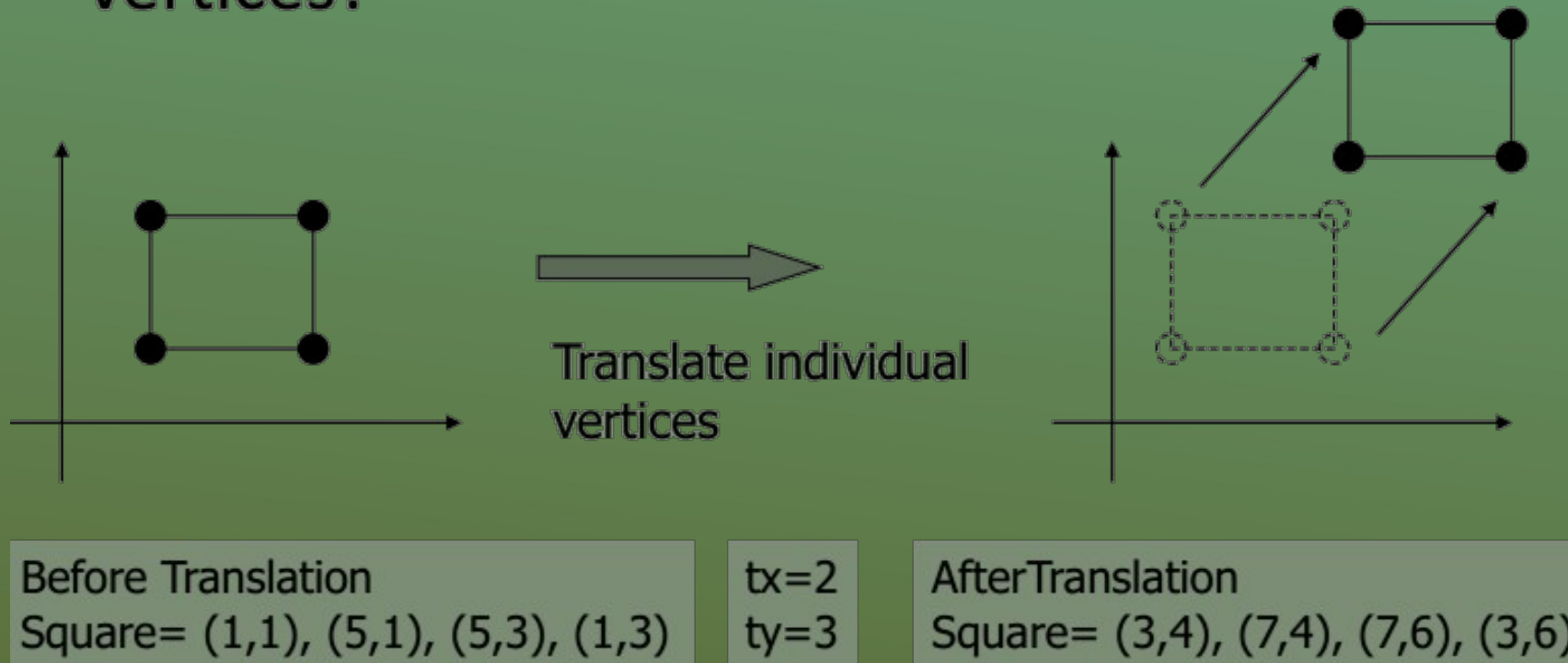
OR $P' = P + T$ where

$$\begin{bmatrix} x' & y' \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix} + \begin{bmatrix} t_x & t_y \end{bmatrix} \quad \text{OR}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

Translation

- How to translate an object with multiple vertices?

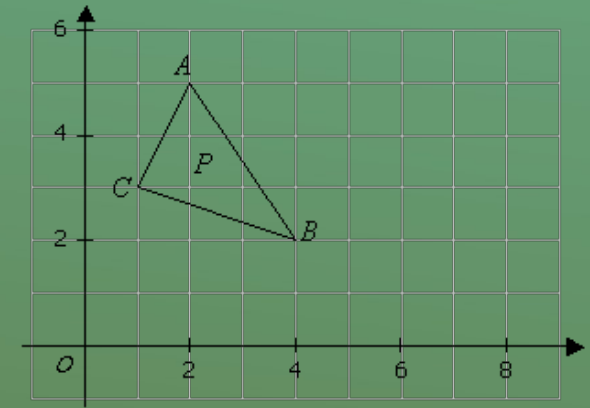


Translation is a process of changing the position of an object in a straight-line path from one coordinate location to another.

Example:

The triangle P is mapped onto the triangle Q by the translation $\begin{pmatrix} 4 \\ -1 \end{pmatrix}$

- a) Find the coordinates of triangle Q .
- b) On the diagram, draw and label triangle Q .

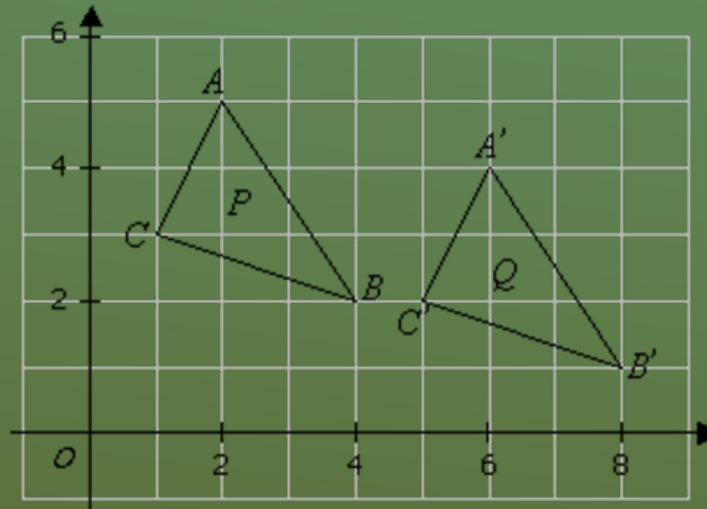


Solution:

a)

$$A' = \begin{pmatrix} 2 \\ 5 \end{pmatrix} + \begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$
$$B' = \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 8 \\ 1 \end{pmatrix}$$
$$C' = \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

b)



Translate a polygon with co-ordinates A(2,5) B(7,10) and C(10,2) by 3 units in X direction and 4 units in Y direction.

$$A' = A + T$$

$$= \begin{bmatrix} 2 \\ 5 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

$$B' = B + T$$

$$= \begin{bmatrix} 7 \\ 10 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 10 \\ 14 \end{bmatrix}$$

$$C' = C + T$$

$$= \begin{bmatrix} 10 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 13 \\ 6 \end{bmatrix}$$

Scaling

- ❖ A scaling transformation changes the size of an object.
- ❖ This operation can be carried out for polygons by multiplying the co-ordinates values (x , y) of each vertex by scaling factors S_x and S_y to produce the transformed co-ordinates (x' , y').

$$x' = x \cdot S_x$$

$$y' = y \cdot S_y$$

- ❖ In the matrix form

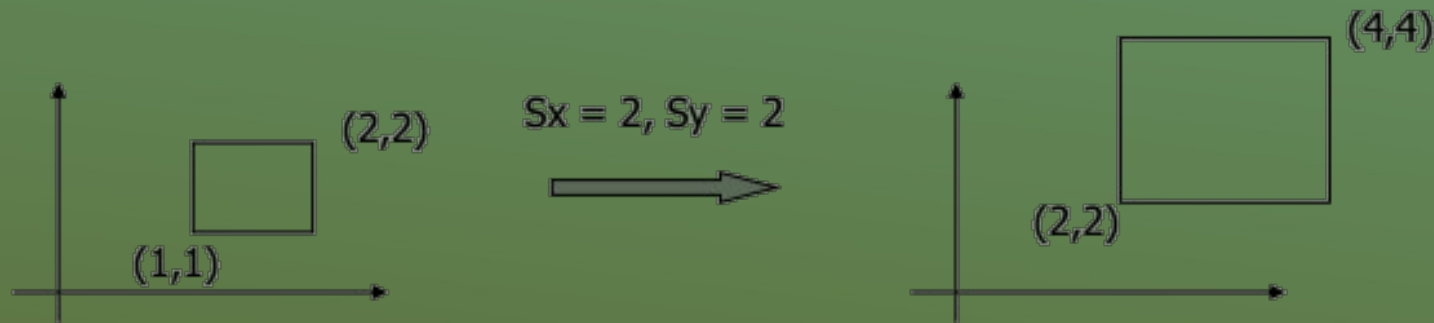
$$\begin{aligned} [x' \quad y'] &= [x \quad y] \begin{pmatrix} S_x & 0 \\ 0 & S_y \end{pmatrix} \\ &= P \cdot S \end{aligned}$$

2D Scaling

Scale: Alter the size of an object by a scaling factor (S_x, S_y) , i.e.

$$\begin{aligned}x' &= x \cdot S_x \\ y' &= y \cdot S_y\end{aligned}$$

$$\Rightarrow \begin{vmatrix} x' \\ y' \end{vmatrix} = \begin{vmatrix} S_x & 0 \\ 0 & S_y \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix}$$



If $1 < S_x \text{ \& } S_y < 1$ then Object size decreases /Point closer to origin

If $S_x \text{ \& } S_y > 1$ then Object size increases/Point away from origin

If $S_x = S_y \rightarrow$ Scaling is done uniformly, shape not changed else shape changes

Scaling

Uniform Scaling



Un-uniform Scaling

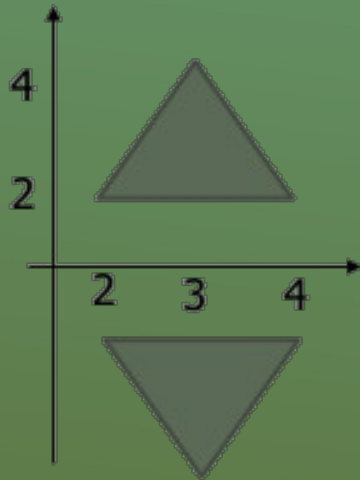


Reflection

i) About x- axis

$$x' = x$$

$$y' = -y$$



Coordinates are (2,2), (4,2), (3,4)

$$(2,2) \rightarrow (2,-2),$$

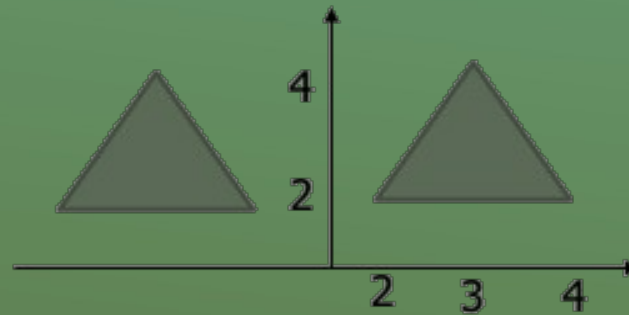
$$(4,2) \rightarrow (4,-2)$$

$$(3,4) \rightarrow (3,-4)$$

i) About y- axis

$$x' = -x$$

$$y' = y$$



Coordinates are (2,2), (4,2), (3,4)

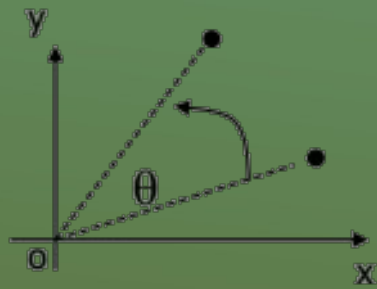
$$(2,2) \rightarrow (-2,2),$$

$$(4,2) \rightarrow (-4,2)$$

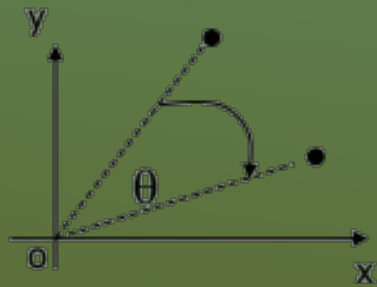
$$(3,4) \rightarrow (-3,4)$$

2D Rotation

Default rotation center: Origin (0,0)



$\theta > 0$: Rotate counter clockwise



$\theta < 0$: Rotate clockwise

2D Rotation

$P(x,y)$ $\rightarrow$ Rotate *about the origin* by θ

$\longrightarrow P'(x', y')$

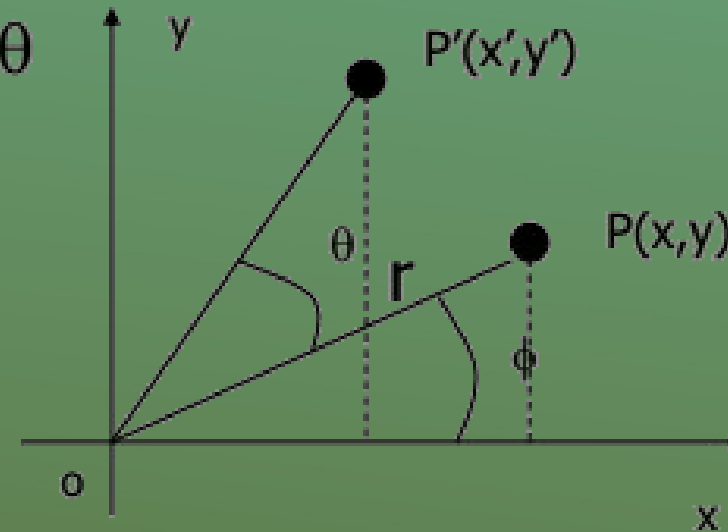
How to compute (x', y') ?

After rotation, new angle = $(\phi + \theta)$

Now

$$x = r \cos(\phi) \quad y = r \sin(\phi)$$

$$x' = r \cos(\phi + \theta) \quad y' = r \sin(\phi + \theta)$$



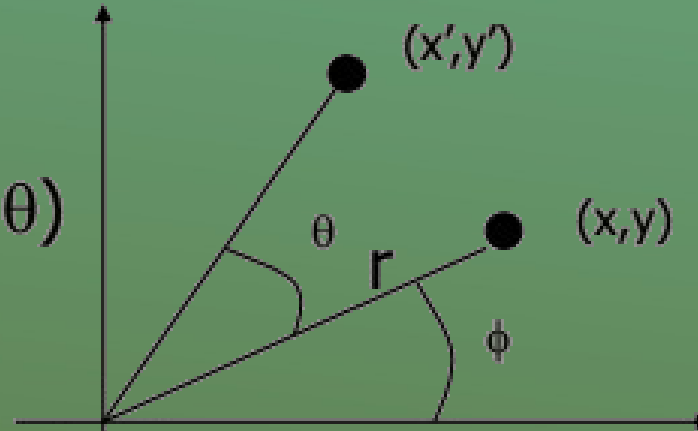
2D Rotation

$$x = r \cos(\phi) \quad y = r \sin(\phi)$$

$$x' = r \cos(\phi + \theta) \quad y' = r \sin(\phi + \theta)$$

$$\begin{aligned} x' &= r \cos(\phi + \theta) \\ &= r \cos(\phi) \cos(\theta) - r \sin(\phi) \sin(\theta) \\ &= x \cos(\theta) - y \sin(\theta) \end{aligned}$$

$$\begin{aligned} y' &= r \sin(\phi + \theta) \\ &= r \sin(\phi) \cos(\theta) + r \cos(\phi) \sin(\theta) \\ &= y \cos(\theta) + x \sin(\theta) \end{aligned}$$



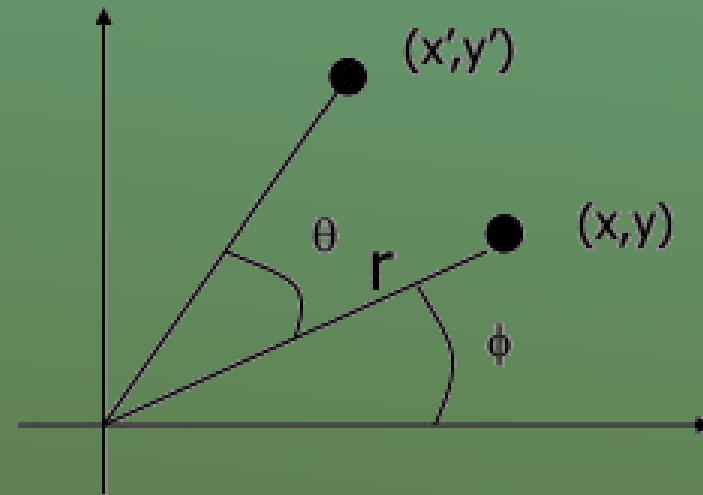
2D Rotation

$$x' = x \cos(\theta) - y \sin(\theta)$$

$$y' = y \cos(\theta) + x \sin(\theta)$$

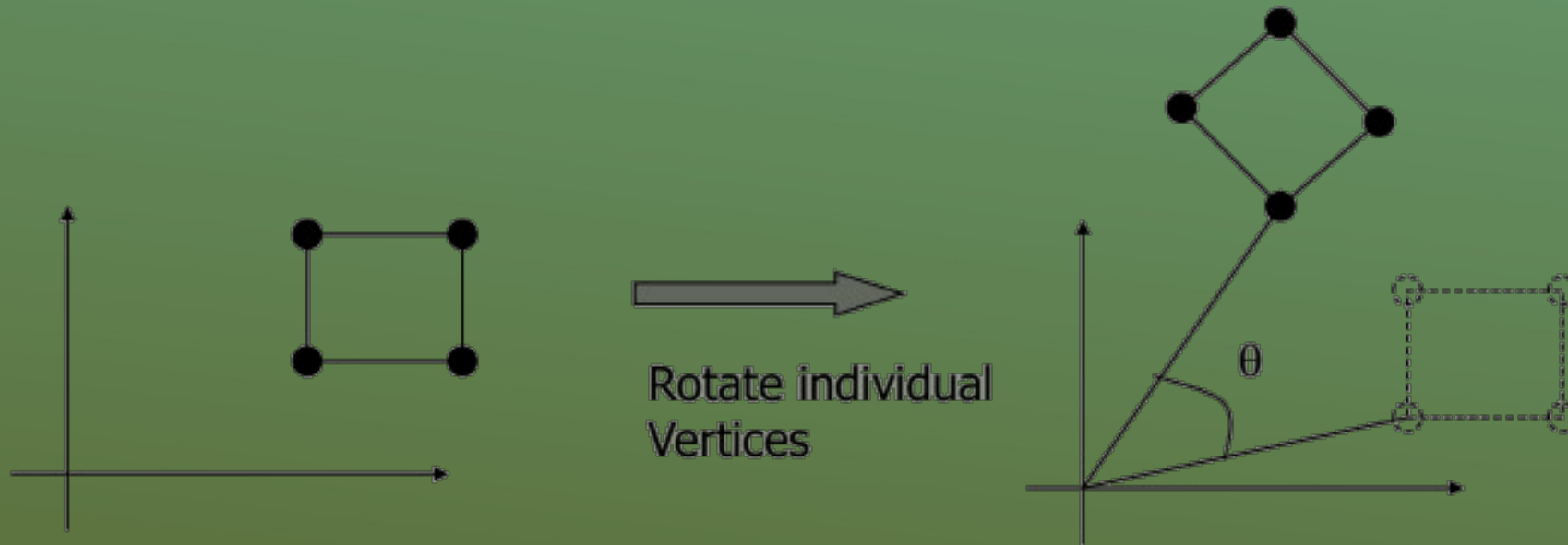
Matrix form?

$$\begin{vmatrix} x' \\ y' \end{vmatrix} = \begin{vmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix}$$



2D Rotation

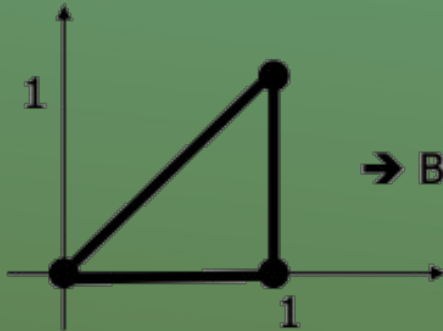
- How to rotate an object with multiple vertices?



2D Rotation

Example: Suppose a Triangle has three sides $(0,0)$, $(1,0)$, $(1,1)$

Let $\theta = 90^\circ$ (anticlockwise)



→ Before Rotation

$$\begin{vmatrix} x' \\ y' \end{vmatrix} = \begin{vmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix}$$

For $(0,0)$

$$\begin{vmatrix} x' \\ y' \end{vmatrix} = \begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix} \begin{vmatrix} 0 \\ 0 \end{vmatrix}$$

→ $(x', y') = (0,0)$

For $(1,0)$

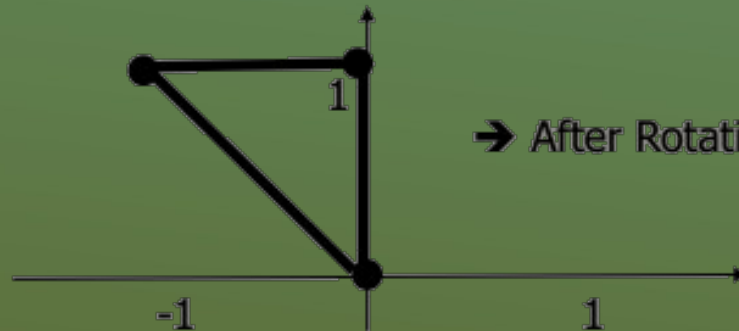
$$\begin{vmatrix} x' \\ y' \end{vmatrix} = \begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix} \begin{vmatrix} 1 \\ 0 \end{vmatrix}$$

→ $(x', y') = (0,1)$

For $(1,1)$

$$\begin{vmatrix} x' \\ y' \end{vmatrix} = \begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix} \begin{vmatrix} 1 \\ 1 \end{vmatrix}$$

→ $(x', y') = (-1,1)$



→ After Rotation

A point (4,3) is rotated counterclockwise by angle of 45.
find the rotation matrix and the resultant point.

$$R = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \cos 45 & -\sin 45 \\ \sin 45 & \cos 45 \end{pmatrix}$$

$$= \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$

$$P' = \begin{pmatrix} 4 & 3 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 4/\sqrt{2} - 3/\sqrt{2} & 4/\sqrt{2} + 3/\sqrt{2} \end{pmatrix}$$

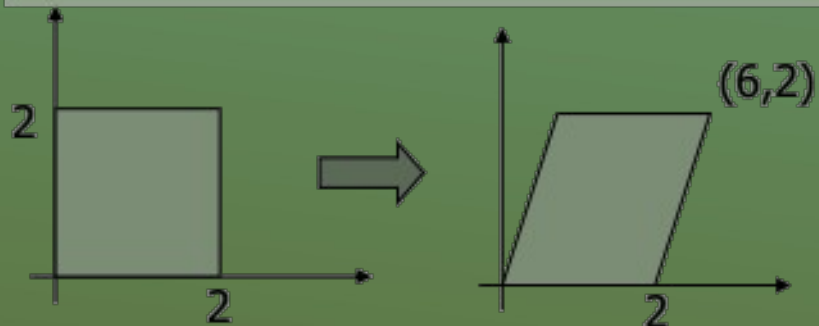
$$= \begin{pmatrix} 1/\sqrt{2} & 7/\sqrt{2} \end{pmatrix}$$

Shearing

x- shear ← skewing
y- share

x- shearing
 $y' = y$
 $x' = x + shr*y$

Y coordinates are unaffected, but x coordinates are translated linearly with y

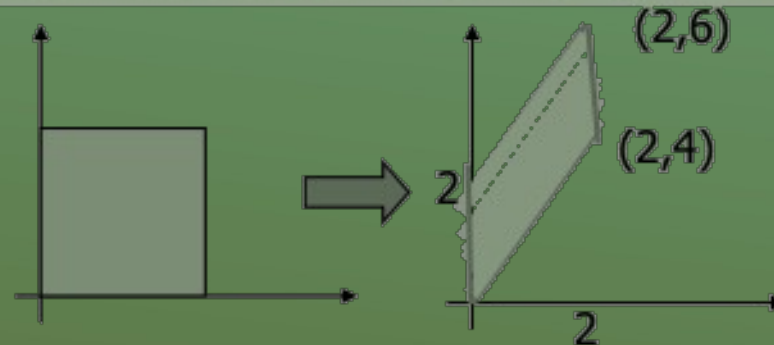


Coordinates are (0,0), (0,2), (2,2), (2,0)

$$\begin{aligned}(0,0) &\rightarrow (0,0), \\ (0,2) &\rightarrow (0+2*2, 2)=(4,2) \\ (2,0) &\rightarrow (2+2*0, 0)=(2,0) \\ (2,2) &\rightarrow (2+2*2,2)=(6,2)\end{aligned}$$

y- shearing
 $x' = x$
 $y' = y + shr*x$

x coordinates are unaffected, but y coordinates are translated linearly with x



Coordinates are (0,0), (0,2), (2,2), (2,0)

$$\begin{aligned}(0,0) &\rightarrow (0,0), \\ (0,2) &\rightarrow (0, 2+2*0)=(0,2) \\ (2,0) &\rightarrow (2, 0+2*2)=(2,4) \\ (2,2) &\rightarrow (2, 2+2*2)=(2,6)\end{aligned}$$